

Boundary Localization of $M_{-1,1}$, and the General Boundary Version

Discussion note

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0. Setup

For $A_1^{(1)}$, let $\widehat{\mathfrak{sl}}_2 = \widehat{\mathfrak{sl}}_2$, with

$$[e_i, f_j] = h_{i+j} + i\delta_{i+j,0}K, \quad [h_i, e_j] = 2e_{i+j}, \quad [h_i, f_j] = -2f_{i+j}, \quad [h_i, h_j] = 2i\delta_{i+j,0}K.$$

For integers N_1, N_2 , put

$$S_{N_1, N_2} = \text{span}\{K, h(n_0), e(n_1), f(n_2) : n_0 \geq 0, n_1 > N_1, n_2 > N_2\}.$$

In particular, $h_0 \in S_{0,0}$. Let

$$M = M_{N_1, N_2}(\varphi) = \text{Ind}_{S_{N_1, N_2}}^{\widehat{\mathfrak{sl}}_2} \mathbb{C}_\varphi, \quad u = 1 \otimes 1,$$

and set $c = \varphi(K)$, $\lambda_i = \varphi(h_i)$. Then

$$e_i u = 0 \quad (i > N_1), \quad f_j u = 0 \quad (j > N_2),$$

$$K u = c u, \quad h_i u = \lambda_i u \quad (i \geq 0).$$

Since φ is a character, $[e_i, f_j] = h_{i+j} + i\delta_{i+j,0}K$ with $i > N_1, j > N_2$ gives

$$\lambda_i = 0 \quad (i \geq N_1 + N_2 + 2).$$

Let Γ_m be the set of finite nondecreasing sequences

$$\gamma = (m_1, \dots, m_k), \quad m_i \leq m, \quad m_1 \leq \dots \leq m_k,$$

and write $x(\gamma) = x(m_1) \cdots x(m_k)$, $x(\emptyset) = 1$, for $x = e, h, f$.

Proposition 1 (PBW basis). *A PBW basis of $M_{N_1, N_2}(\varphi)$ is*

$$\mathcal{B}_{N_1, N_2} = \{e(\alpha)h(\beta)f(\gamma)u : \alpha \in \Gamma_{N_1}, \beta \in \Gamma_{-1}, \gamma \in \Gamma_{N_2}\}.$$

Proof. Take the vector space complement

$$\mathfrak{a} = \text{span}\{e_i : i \leq N_1\} \oplus \text{span}\{h_i : i \leq -1\} \oplus \text{span}\{f_i : i \leq N_2\}$$

of S_{N_1, N_2} in $\widehat{\mathfrak{sl}}_2$. By the PBW theorem,

$$U(\widehat{\mathfrak{sl}}_2) \cong U(\mathfrak{a}) \otimes U(S_{N_1, N_2})$$

as vector spaces. Tensoring with the one-dimensional S_{N_1, N_2} -module $\mathbb{C}_\varphi$ turns the S_{N_1, N_2} -factors into scalars, leaving exactly the ordered PBW monomials of $\mathfrak{a}$. $\square$

Lemma 2 (Localization formulas). *For $F = f_b$ and $s \in \mathbb{Z}$,*

$$\begin{aligned} f_n F^s &= F^s f_n, \\ h_n F^s &= F^s h_n - 2s F^{s-1} f_{n+b}, \\ e_n F^s &= F^s e_n + s F^{s-1} (h_{n+b} + n \delta_{n+b,0} K) - 2 \binom{s}{2} F^{s-2} f_{n+2b}. \end{aligned}$$

Proof. One has

$$\begin{aligned} [F, f_n] &= 0, & [F, h_n] &= 2f_{n+b}, & [F, e_n] &= -(h_{n+b} + n \delta_{n+b,0} K), \\ (\text{ad } F)^2(e_n) &= -2f_{n+2b}, & (\text{ad } F)^3(e_n) &= 0. \end{aligned}$$

In the localization,

$$x F^s = \sum_{j \geq 0} (-1)^j \binom{s}{j} F^{s-j} (\text{ad } F)^j(x),$$

a finite sum. □

1. The case $N_1 = -1, N_2 = 1$

Fix

$$M = M_{-1,1}(\varphi), \quad F = f_1.$$

Since F is a PBW variable, F acts injectively on M . Let

$$\mathcal{D}_F M = \{F^m : m \geq 0\}^{-1} M.$$

Because the f_i commute pairwise, $\mathcal{D}_F M$ has basis

$$\{F^s e(\alpha) h(\beta) f(\delta) u : s \in \mathbb{Z}, \alpha, \beta \in \Gamma_{-1}, \delta \in \Gamma_0\},$$

and M corresponds to $s \geq 0$. The localization quotient

$$T_F M = \mathcal{D}_F M / M$$

has PBW basis

$$\mathcal{C} = \{F^{-m} e(\alpha) h(\beta) f(\delta) u + M : m \geq 1, \alpha, \beta \in \Gamma_{-1}, \delta \in \Gamma_0\}.$$

Put

$$\bar{u}_m = F^{-m} u + M \quad (m \geq 1), \quad \bar{u} = \bar{u}_1.$$

Define $\psi : S_{0,0} \rightarrow \mathbb{C}$ by

$$\begin{aligned} \psi(K) &= c, & \psi(h_0) &= \lambda_0 + 2, & \psi(h_1) &= \lambda_1, & \psi(h_i) &= 0 \quad (i \geq 2), \\ \psi(e_i) &= 0 \quad (i > 0), & \psi(f_i) &= 0 \quad (i > 0), \end{aligned}$$

and let

$$A = M_{0,0}(\psi) = \text{Ind}_{S_{0,0}}^{\widehat{\text{sl}}_2} \mathbb{C}_\psi, \quad u' = 1 \otimes 1.$$

Lemma 3. *The vector $\bar{u} \in T_F M$ satisfies*

$$x \bar{u} = \psi(x) \bar{u} \quad (x \in S_{0,0}).$$

Proof. For $s = -1$, Lemma 2 gives

$$\begin{aligned} f_n F^{-1} &= F^{-1} f_n, \\ h_n F^{-1} &= F^{-1} h_n + 2F^{-2} f_{n+1}, \\ e_n F^{-1} &= F^{-1} e_n - F^{-2} (h_{n+1} + n\delta_{n+1,0} K) - 2F^{-3} f_{n+2}. \end{aligned}$$

Hence, for $n > 0$,

$$f_n \bar{u} = 0, \quad e_n \bar{u} = 0$$

(for $n = 1$, $f_1 \bar{u} = F^{-1} f_1 u + M = F^{-1} F u + M = u + M = 0$). For the Cartan part,

$$h_0 \bar{u} = (F^{-1} h_0 + 2F^{-2} f_1) u + M = (\lambda_0 + 2) F^{-1} u + M = (\lambda_0 + 2) \bar{u} = \psi(h_0) \bar{u},$$

$$h_1 \bar{u} = (F^{-1} h_1 + 2F^{-2} f_2) u + M = \lambda_1 \bar{u} = \psi(h_1) \bar{u},$$

and, for $n \geq 2$,

$$h_n \bar{u} = (F^{-1} h_n + 2F^{-2} f_{n+1}) u + M = 0 = \psi(h_n) \bar{u}.$$

Also $K \bar{u} = c \bar{u} = \psi(K) \bar{u}$. □

Proposition 4. *There is a unique $\widehat{\mathfrak{sl}}_2$ -module homomorphism*

$$\Phi : A \longrightarrow T_F M, \quad \Phi(u') = \bar{u}.$$

If $\lambda_1 \neq 0$, then Φ is surjective.

Proof. Let

$$\psi_U : U(S_{0,0}) \longrightarrow \mathbb{C}$$

be the algebra character extending the Lie algebra character ψ . By Lemma 3, for every $a \in U(S_{0,0})$ we have

$$a \bar{u} = \psi_U(a) \bar{u}.$$

This relation is not only a relation for the generators in $S_{0,0}$: if $a = s_1 \cdots s_k$, with $s_i \in S_{0,0}$, then $a \bar{u} = \psi(s_1) \cdots \psi(s_k) \bar{u} = \psi_U(a) \bar{u}$, and the general case follows by linearity.

Write $\mathbb{C}_\psi = \mathbb{C} \mathbf{1}_\psi$. On the unbalanced tensor product define

$$\tilde{\Phi} : U(\widehat{\mathfrak{sl}}_2) \otimes \mathbb{C}_\psi \longrightarrow T_F M, \quad \tilde{\Phi}(x \otimes b \mathbf{1}_\psi) = b x \bar{u}.$$

For $x \in U(\widehat{\mathfrak{sl}}_2)$, $a \in U(S_{0,0})$, and $b \in \mathbb{C}$, the balancing relation is sent to

$$\begin{aligned} &\tilde{\Phi}(xa \otimes b \mathbf{1}_\psi - x \otimes a(b \mathbf{1}_\psi)) \\ &= b x a \bar{u} - b \psi_U(a) x \bar{u} = b x (a \bar{u} - \psi_U(a) \bar{u}) = 0. \end{aligned}$$

Hence $\tilde{\Phi}$ kills the balancing subspace and descends to the well-defined linear map

$$\Phi : A = U(\widehat{\mathfrak{sl}}_2) \otimes_{U(S_{0,0})} \mathbb{C}_\psi \longrightarrow T_F M, \quad \Phi(x \otimes_{U(S_{0,0})} b \mathbf{1}_\psi) = b x \bar{u}.$$

For $y \in \widehat{\mathfrak{sl}}_2$,

$$\Phi(y \cdot (x \otimes_{U(S_{0,0})} b \mathbf{1}_\psi)) = b y x \bar{u} = y \cdot (b x \bar{u}),$$

so Φ is an $\widehat{\mathfrak{sl}}_2$ -module homomorphism. In particular, $\Phi(u') = \bar{u}$; since $A = U(\widehat{\mathfrak{sl}}_2) u'$, this condition also gives uniqueness.

For $m \geq 1$,

$$e_0 F^{-m} = F^{-m} e_0 - m F^{-m-1} h_1 - m(m+1) F^{-m-2} f_2.$$

Thus

$$e_0 \bar{u}_m = -m\lambda_1 \bar{u}_{m+1},$$

and

$$e_0^r \bar{u} = (-1)^r r! \lambda_1^r \bar{u}_{r+1} \quad (r \geq 0).$$

If $\lambda_1 \neq 0$, then

$$\bar{u}_m = \frac{(-1)^{m-1}}{(m-1)! \lambda_1^{m-1}} e_0^{m-1} \bar{u} \in \text{Im } \Phi.$$

A PBW basis of $T_F M$ is

$$B_{m,\eta} := F^{-m} X_\eta u + M, \quad X_\eta = e(\alpha) h(\beta) f(\delta),$$

where $m \geq 1$, $\alpha, \beta \in \Gamma_{-1}$, $\delta \in \Gamma_0$. Fix a total PBW monomial order $\prec$ on the exterior multi-indices η , for instance the degree-lexicographic order induced by the chosen PBW ordering of the generators. For

$$C_{m,\eta} = \frac{(-1)^{m-1}}{(m-1)! \lambda_1^{m-1}} X_\eta e_0^{m-1} u' \in A,$$

one has

$$\Phi(C_{m,\eta}) = X_\eta \bar{u}_m.$$

Moving the exterior factors in X_η past F^{-m} gives a finite triangular expansion

$$\Phi(C_{m,\eta}) = B_{m,\eta} + \sum_{\eta' \prec \eta, m'} a_{m',\eta'} B_{m',\eta'} + \sum_{i=0}^{N_\eta} r_i(m, \eta) \bar{u}_{m+i},$$

where $a_{m',\eta'}, r_i(m, \eta) \in \mathbb{C}$; the last finite sum is precisely the part with zero exterior PBW index. We prove $B_{m,\eta} \in \text{Im } \Phi$ by induction on η . The case $\eta = 0$ is $\bar{u}_m \in \text{Im } \Phi$. Assuming

$$B_{m',\eta'} \in \text{Im } \Phi \quad (\eta' \prec \eta, m' \geq 1),$$

the triangular formula gives

$$B_{m,\eta} = \Phi(C_{m,\eta}) - \sum_{\eta' \prec \eta, m'} a_{m',\eta'} B_{m',\eta'} - \sum_{i=0}^{N_\eta} r_i(m, \eta) \bar{u}_{m+i} \in \text{Im } \Phi.$$

Hence every PBW basis vector of $T_F M$ lies in $\text{Im } \Phi$. □

Theorem 5. *Assume*

$$\lambda_1 = \varphi(h_1) \neq 0.$$

Then

$$\Phi : M_{0,0}(\psi) \longrightarrow T_{f_1} M_{-1,1}(\varphi)$$

is an isomorphism.

Proof. By Proposition 4, Φ is surjective. For injectivity, take for $A = M_{0,0}(\psi)$ the PBW basis

$$\mathcal{B}_{0,0} = \{ e_0^p e(\alpha) h(\beta) f(\delta) u' : p \geq 0, \alpha, \beta \in \Gamma_{-1}, \delta \in \Gamma_0 \},$$

and order both $\mathcal{B}_{0,0}$ and $\mathcal{C}$ lexicographically by (p, η) and (m, η) , respectively, with η ordered by the degree-lexicographic order induced by the chosen PBW ordering, under the correspondence

$$e_0^p e(\alpha) h(\beta) f(\delta) u' \longleftrightarrow F^{-p-1} e(\alpha) h(\beta) f(\delta) u + M.$$

By Lemma 2, the leading term of

$$\Phi(e_0^p e(\alpha) h(\beta) f(\delta) u')$$

is

$$(-1)^p p! \lambda_1^p F^{-p-1} e(\alpha) h(\beta) f(\delta) u + M;$$

the remaining terms come from the commutators produced when moving e_0, h_i, e_i, f_i past powers of F , and they either involve lower PBW monomials or lie in lower filtration layers under the same monomial. Thus Φ is upper triangular between the two PBW bases, with diagonal coefficients

$$(-1)^p p! \lambda_1^p \neq 0.$$

Therefore Φ is injective. □

Remark 6. If $\lambda_1 = 0$, the isomorphism generally fails. Then

$$e_0 \bar{u} = 0,$$

while in $M_{0,0}(\psi)$ the element e_0 is a free variable of the PBW complement and cannot be killed by the generating vector.

Theorem 7. *Assume*

$$\lambda_1 = \varphi(h_1) \neq 0, \quad c = \varphi(K) \neq -2.$$

Then

$$T_{f_1} M_{-1,1}(\varphi) = \mathcal{D}_{f_1} M_{-1,1}(\varphi) / M_{-1,1}(\varphi)$$

is a simple $\widehat{\mathfrak{sl}}_2$ -module.

Proof. The FGXZ irreducibility criterion gives simplicity of $M_{N_1, N_2}(\chi)$ when

$$\chi(h(N_1 + N_2 + 1)) \neq 0, \quad \chi(K) \neq -2.$$

For $A = M_{0,0}(\psi)$, this is

$$\psi(h_1) = \lambda_1 \neq 0, \quad \psi(K) = c \neq -2.$$

Thus A is simple, and by Theorem 5 it is isomorphic to $T_{f_1} M_{-1,1}(\varphi)$. Hence the latter is simple. □

2. The general boundary version

Theorem 8. *Let $N \geq 1$,*

$$M = M_{-1,N}(\varphi), \quad F = f_N,$$

and assume

$$\lambda_N = \varphi(h_N) \neq 0.$$

Then

$$\mathcal{D}_{f_N} M_{-1,N}(\varphi) / M_{-1,N}(\varphi) \cong M_{0,N-1}(\psi),$$

where $\psi : S_{0,N-1} \rightarrow \mathbb{C}$ is given by

$$\psi(K) = c, \quad \psi(h_0) = \lambda_0 + 2, \quad \psi(h_i) = \lambda_i \ (i \geq 1).$$

If moreover $c \neq -2$, then the quotient is a simple $\widehat{\mathfrak{sl}}_2$ -module.

Proof. The proof for $N = 1$ carries over verbatim with the shifts $n+1 \rightsquigarrow n+N$, $n+2 \rightsquigarrow n+2N$. Lemma 2 gives, for $s = -1$,

$$\begin{aligned} f_n F^{-1} &= F^{-1} f_n, \\ h_n F^{-1} &= F^{-1} h_n + 2F^{-2} f_{n+N}, \\ e_n F^{-1} &= F^{-1} e_n - F^{-2} (h_{n+N} + n\delta_{n+N,0} K) - 2F^{-3} f_{n+2N}. \end{aligned}$$

Hence $x\bar{u} = \psi(x)\bar{u}$ for $x \in S_{0,N-1}$: for $n = N$ one uses

$$f_N \bar{u} = F^{-1} f_N u + M = F^{-1} F u + M = u + M = 0,$$

the Cartan and e -remainders are killed by $\lambda_i = 0$ ($i > N$), and $h_0 \bar{u} = (\lambda_0 + 2)\bar{u}$ since $2F^{-2} f_N u = 2F^{-2} F u = 2\bar{u}$. Moreover

$$e_0 F^{-m} = F^{-m} e_0 - mF^{-m-1} h_N - m(m+1)F^{-m-2} f_{2N},$$

so

$$e_0 \bar{u}_m = -m\lambda_N \bar{u}_{m+1}$$

(because $f_{2N} u = 0$), and $e_0^r \bar{u} = (-1)^r r! \lambda_N^r \bar{u}_{r+1}$. The PBW bases of the quotient and of $M_{0,N-1}(\psi)$ are

$$\{F^{-m} e(\alpha) h(\beta) f(\delta) u + M : m \geq 1, \alpha, \beta \in \Gamma_{-1}, \delta \in \Gamma_{N-1}\}$$

and

$$\{e_0^p e(\alpha) h(\beta) f(\delta) u' : p \geq 0, \alpha, \beta \in \Gamma_{-1}, \delta \in \Gamma_{N-1}\},$$

and the triangular argument of Theorem 5 yields the isomorphism, the diagonal coefficients being

$$(-1)^p p! \lambda_N^p \neq 0.$$

Simplicity follows from the FGXZ criterion applied to $M_{0,N-1}(\psi)$, which requires

$$\psi(h_N) = \lambda_N \neq 0, \quad \psi(K) = c \neq -2.$$

□

Remark 9 (General (a, b)). For $M_{a,b}(\varphi)$ localized at the boundary f_b , assuming

$$\varphi(h_{a+b+1}) \neq 0,$$

the same triangular PBW argument is expected to give

$$\mathcal{D}_{f_b} M_{a,b}(\varphi) / M_{a,b}(\varphi) \cong M_{a+1,b-1}(\psi),$$

where $\psi : S_{a+1,b-1} \rightarrow \mathbb{C}$ is given by

$$\psi(K) = c, \quad \psi(h_0) = \lambda_0 + 2, \quad \psi(h_i) = \lambda_i \quad (i \geq 1),$$

$$\psi(e_i) = 0 \quad (i > a+1), \quad \psi(f_j) = 0 \quad (j > b-1).$$

By the FGXZ criterion, $M_{a+1,b-1}(\psi)$ is simple when

$$\psi(h_{a+b+1}) = \varphi(h_{a+b+1}) \neq 0, \quad \psi(K) = c \neq -2,$$

in which case the localization quotient is simple.